

Probability as a measure

Outline

- 1) What is a probability?
- 2) Measures and integrals
- 3) Densities
- 4) Probability spaces

What is probability?

Two common answers:

- 1) Frequentist answer: Relative frequency over many repetitions of a given experiment.
- 2) Bayesian answer: Degree of belief that something is true, or will happen.

Many disagreements, e.g. about what kinds of things we can meaningfully assign prob. to.

- Prob. (Die roll lands on 4)
- Prob. (Surgery is successful)
- Prob. (Harris wins upcoming election)
- Prob. (Subatomic particle has predicted mass)
- Prob. ($P = NP$)
- Prob. (20th digit of $\sqrt{2}$ is 5)

Bayesians get mileage by putting prob.s on everything.

Frequentists try to avoid this.

Many controversies about this!

Mathematical Probability

Fortunately, these disagreements don't extend to the mathematical construct of probability

Mathematical answer: A function P mapping (some) subsets of a sample space $\mathcal{X}$ to $[0, 1]$, which is additive over disjoint sets:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad \text{if } A_i \cap A_j = \emptyset, \text{ all } i \neq j$$

and has $P(\mathcal{X}) = 1$

Originally invented to analyze games of chance

Classical theory for discrete events defined by Laplace

Problems remained for continuous variables, e.g.

- treatment of pathological sets
- conditioning on probability zero events

Kolmogorov (1933) recognized:

- probability is a special case of a measure,
- expectation is an integral against a prob. measure

Measure theory basics

Measure theory is a rigorous grounding
for probability theory [subject of 205A]

Simplifies notation & clarifies concepts, especially
around integration & conditioning

Given a set X , a measure μ maps subsets
 $A \subseteq X$ to non-negative numbers $\mu(A) \in [0, \infty]$

Example X countable (e.g. $X = \mathbb{Z}$)

Counting measure $\#(A) = \# \text{ points in } A$

Example $X = \mathbb{R}^n$

Lebesgue measure $\lambda(A) = \int_A \dots \int dx_1 \dots dx_n$
 $= \text{Volume}(A)$

Standard Gaussian distribution:

$$\begin{aligned} P_z(A) &= \mathbb{P}(Z \in A) \quad \text{where } Z \sim N(0, 1) \\ &= \int_A \phi(x) dx \quad \phi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}} \end{aligned}$$

NB Because of pathological sets, $\lambda(A)$ can only
be defined for certain subsets $A \subseteq \mathbb{R}^n$

In general, the domain of a measure μ is a collection of subsets $\mathcal{F} \subseteq 2^X$ (power set)

$\mathcal{F}$ should satisfy certain closure properties

(technical term: σ -field)

(not important for us)

$$\textcircled{1} X \in \mathcal{F}$$

$$\textcircled{2} \text{ If } A \in \mathcal{F} \text{ then } X \setminus A \in \mathcal{F}$$

$$\textcircled{3} \text{ If } A_1, A_2, \dots \in \mathcal{F} \text{ then } \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$$

Ex: X countable, $\mathcal{F} = 2^X$

Ex: $X = \mathbb{R}^n$, $\mathcal{F} = \text{Borel } \sigma\text{-field } \mathcal{B}$

$\mathcal{B} = \text{smallest } \sigma\text{-field including all open rectangles}$
 $(a_1, b_1) \times \dots \times (a_n, b_n) \quad a_i < b_i \quad \forall i$

Given a measurable space $(X, \mathcal{F})$ a measure is a map $\mu: \mathcal{F} \rightarrow [0, \infty]$ with

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i) \quad \text{for disjoint } A_1, A_2, \dots \in \mathcal{F}$$

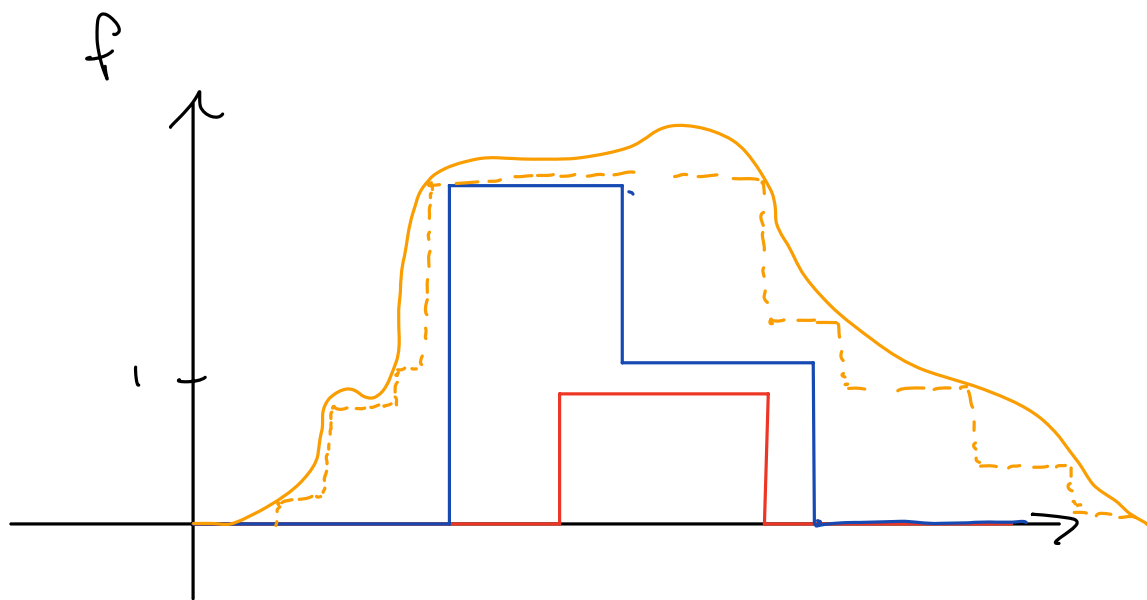
$$\mu(\emptyset) = 0$$

μ probability measure if $\mu(X) = 1$

Integrals

Measures let us define integrals that put weight $\mu(A)$ on $A \subseteq X$

Define $\int 1_{\{x \in A\}} d\mu(x) = \mu(A)$, extend to other functions by linearity & limits:



Indicator $\int 1_{\{x \in A\}} d\mu(x) = \mu(A)$

Simple Function $\int \left(\sum c_i 1_{\{x \in A_i\}} \right) d\mu(x) = \sum c_i \mu(A_i)$

"Nice enough" (measurable) + function $\int f(x) d\mu(x)$ approximated by simple functions

Examples:

Counting: $\int f d\# = \sum_{x \in X} f(x)$

↙ Lebesgue
integral

Lebesgue: $\int f d\lambda = \int \dots \int f(x) dx_1 \dots dx_n$

Gaussian: Note $\int 1_A(x) dP_z(x) = P_z(A) = \int_{-\infty}^{\infty} 1_A \phi dx$

By extension,

$$\int f dP_z = \int f(x) \phi(x) dx = \mathbb{E}[f(z)]$$

To evaluate $\int f dP_z$ rewrite as $\int f \phi dx$.
↙ density [can't always
do this]
e.g. Binom

It is nice to turn integrals we care
about into Lebesgue integrals. When
can we do this?

Densities

λ and P above are closely related. Want to make this precise.

Given $(X, \mathcal{F})$, two measures P, μ

We say P is absolutely continuous wrt μ
if $P(A) = 0$ whenever $\mu(A) = 0$

Notation: $P \ll \mu$ or we say μ dominates P

If $P \ll \mu$ then (under mild conditions) we can always define a density function

$p: X \rightarrow [0, \infty)$ with

$$P(A) = \int_A p(x) d\mu(x)$$

$$\int f(x) dP(x) = \int f(x) p(x) d\mu(x)$$

Sometimes written $p(x) = \frac{dP}{d\mu}(x)$, called
Radon - Nikodym derivative

Densities are very useful:

Turn $\int f(x) dP(x)$ into something we know how to evaluate, such as

$$1) \int_{\mathcal{X}} f(x) p(x) dx \quad (X \text{ continuous, } \mathcal{X} \subseteq \mathbb{R}^n)$$

$p(x)$ called probability density function (pdf)

$$2) \sum_{x \in \mathcal{X}} f(x) p(x) \quad (X \text{ discrete, } \mathcal{X} \text{ countable})$$

$p(x)$ called probability mass function (pmf)

Often define distributions by giving their density wrt some known measure, e.g.

Ex: Binom (n, θ) pmf: $p(x) = \binom{n}{x}, x = 0, \dots, n$

(density p wrt counting measure on $\mathcal{X} = \{0, \dots, n\}$)

Note this dist. has no density wrt Lebesgue:

$$\int_{\{0, \dots, n\}} p(x) dx = 0 \quad \text{for any function } p$$

Probability space, random variables

Problem setup may have many random outcomes with complex relationships to one another

Convenient to start with abstract outcome $\omega \in \Omega$

- represents "everything that happens"
- quantities of interest are functions of ω

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space

$\omega \in \Omega$ called outcome

$A \in \mathcal{F}$ called event

$\mathbb{P}(A)$ called probability of A

A random variable is a function $X: \Omega \rightarrow \mathcal{X}$

We say X has distribution Q ($X \sim Q$)

$$\begin{aligned} \text{if } \mathbb{P}(X \in B) &= \mathbb{P}(\{\omega: X(\omega) \in B\}) \\ &= Q(B) \end{aligned}$$

$Q(B)$ is the push-forward of $\mathbb{P}$: $Q(B) = \mathbb{P} \circ X^{-1}(B)$

Idea applies more generally: if μ measure on $\mathcal{X}$,

$f: \mathcal{X} \rightarrow \mathcal{Y}$ induces new measure $\nu(B) = \mu(f^{-1}(B))$

Can write events involving many R.V.s:

$$\mathbb{P}(X > Y > Z \geq 0) = \mathbb{P}(\{\omega: \dots\})$$

The expectation is an integral w.r.t. $\mathbb{P}$

$$\mathbb{E}[f(X, Y)] = \int_{\Omega} f(X(\omega), Y(\omega)) d\mathbb{P}(\omega)$$

To do real calculations we must eventually boil
 $\mathbb{P}$ or $\mathbb{E}$ down to concrete integrals/sums/etc.

If $\mathbb{P}(A) = 1$ we say A occurs almost surely

More in Keener ch. 1, much more in Stat 205A